

COMPACTNESS IN THE LATTICE OF SUBGROUPS OF A TOPOLOGICAL GROUP

Yu. A. Komarov and I. V. Protasov

UDC 519.46

The study of topological groups with respect to the properties of systems of their closed subgroups is one of the main directions of the algebraic theory of topological groups. In this connection, as the defining restrictions, as a rule we choose suitable analogs of restrictions from the domain of discrete groups. Basing on the topology of a group G , we can introduce a topology in the lattice $\mathfrak{L}(G)$ of its closed subgroups, which, in its turn, opens up new possibilities for the study of groups with respect to the properties of the topologized systems of their closed subgroups. Since even in the case of a discrete group G the topology in $\mathfrak{L}(G)$ is, in general, nontrivial, in this way there arise new results in the theory of discrete groups also.

Let G be a topological group, $\mathfrak{L}_E(G)$ be the lattice of its closed subgroups that is equipped with the E -topology (see [1]). The structure of locally compact groups with a compact lattice $\mathfrak{L}_E(G)$ (the EK -groups) has been studied in [1]. Let us denote the families of the monothetic, the Abelian, and the normal subgroups from $\mathfrak{L}_E(G)$ by $\mathfrak{M}(G)$, $\mathfrak{A}(G)$, and $\mathfrak{N}(G)$, respectively, and the families of the compact and the noncompact subgroups from $\mathfrak{L}_E(G)$ by $\mathfrak{K}(G)$ and $n\mathfrak{K}(G)$, respectively.

We retain the terminology and the notation of [1]; in particular, in the sequel only closed subgroups of a topological group will be called its subgroups. All the considered topological groups, unless stated otherwise, are supposed to be locally compact.

1. Compact Subsets of $\mathfrak{L}_E(G)$

LEMMA 1. Let $\mathfrak{F}$ be a compact set of compact subgroups from $\mathfrak{L}_E(G)$. Then all subgroups from $\mathfrak{F}$ are contained in a compact subset F of G .

Proof. Let U be a compact neighborhood of the identity. Since $\mathfrak{F}$ is compact, the covering $\{D_1(HU), H \in \mathfrak{F}\}$ contains a finite subcovering $\{D_1(H_1U), \dots, D_1(H_nU)\}$. Then each subgroup from $\mathfrak{F}$ is contained in $F = \bigcup_{k=1}^n H_kU$.

LEMMA 2. Let a noncompact subgroup H be a limit point of a subset $\mathfrak{F}$ of $\mathfrak{L}_E(G)$. If $\mathfrak{F}$ does not contain any infinite discrete closed subset, then H is a limit point of the set of subgroups from $\mathfrak{F}$ that are contained in H .

Proof. Assuming that no subgroup from $\mathfrak{L}$ is contained in H , we construct an infinite discrete closed subset of $\mathfrak{F}$.

Let us observe that for each neighborhood $\mathfrak{K}$ of the subgroup H the set $F = \bigcup_{x \in \mathfrak{F} \cap \mathfrak{K}} (x \setminus H)$ is noncompact.

In the contrary case, we choose an $h \in H$ and a neighborhood V of the identity such that $F \cap Vh = \emptyset$ and $F \cap FVh = \emptyset$. Let $X \in \mathfrak{F} \cap \mathfrak{K} \cap D_2(Vh)$ and $x \in X \cap Vh$. Since $F \cap Vh = \emptyset$, it follows that $x \in H$. If $z \in X \setminus H$, then $zx \in FVh$, and, on the other hand, $zx \in X \setminus H \subseteq F$, which contradicts the relation $F \cap FVh = \emptyset$.

Let U be a compact neighborhood of the identity, H_1 be an arbitrary subgroup from $\mathfrak{F}$, and $x_1 \in H_1 \setminus H$. We take a compact neighborhood V_1 of x_1 that is disjoint from H . Since $H \subseteq G \setminus V_1$, H is a limit point of the subset $\mathfrak{F}_1 = \mathfrak{F} \cap D_1(G \setminus V_1)$. Since $\bigcup (X \setminus H)$ is not compact, we can choose H_2 from $\mathfrak{F}_1$ such that $x_2U \cap x_1U = \emptyset$ for a certain $x_2 \in H_2$. Let V_2 be a compact neighborhood of x_2 such that $V_2 \cap H = \emptyset$. Let us set $\mathfrak{F}_2 = \mathfrak{F}_1 \cap D_1(G \setminus V_2)$ and choose $H_3 \in \mathfrak{F}_2$ such that $x_3U \cap x_1U = \emptyset$ and $x_3U \cap x_2U = \emptyset$ for a certain $x_3 \in H_3$. Continuing this process, we construct a sequence of subgroups $\{H_i\}$, a sequence of elements $\{x_i\}$, and a sequence of their neighborhoods $\{V_i\}$ such that $x_i \in H_i \setminus H$ and $V_i \cap H = \emptyset$ for each natural number i and $H_k \not\subseteq D_2(V_i)$ for $k > i$ and $x_iU \cap x_jU = \emptyset$ for $i \neq j$.

Kiev State University. Translated from *Ukrainskii Matematicheskii Zhurnal*, Vol. 33, No. 2, pp. 184-189, March-April, 1981. Original article submitted June 15, 1979.

Let us set $\mathfrak{H} = \{H_1, \dots, H_n, \dots\}$. Since the neighborhood $D_2(V_n)$ of the group H_n contains only a finite number of subgroups from $\mathfrak{H}$, it follows that $\mathfrak{H}$ is discrete. Suppose that $K \in \mathfrak{H}$, but $K \notin \mathfrak{H}$. Since $H_k \not\subseteq D_2(V_i)$ for any $k > i$, it follows that $x_i \notin K$, $i = 1, 2, \dots$. By construction, the set $\{x_1, \dots, x_n, \dots\}$ is closed; however, the neighborhood $D_1(G \setminus \{x_1, \dots, x_n, \dots\})$ of the subgroup K does not contain any subgroup from $\mathfrak{H}$. Therefore $\mathfrak{H} = \mathfrak{H}$, and the lemma is proved.

COROLLARY 1. If a noncompact subgroup H is a limit point of a compact subset $\mathfrak{F}$ of $\mathfrak{L}_E(G)$, then it is a limit point of the set of those subgroups from $\mathfrak{F}$ that are contained in it.

COROLLARY 2. Let $\mathfrak{F}$ be a linearly ordered (by inclusion) set of noncompact subgroups of G . If $\mathfrak{F}$ is contained in a compact subset of $\mathfrak{L}_E(G)$, then it is well-ordered. In particular, G satisfies the minimality condition for noncompact subgroups from a compact subset $\mathfrak{F}$ of $\mathfrak{L}_E(G)$.

Proof. Let $\mathfrak{F}_1$ be an arbitrary subset of $\mathfrak{F}$. We number the subgroups of $\mathfrak{F}_1$ by indices from a certain set J and assume that $H = \bigcap_{\alpha \in J} H_\alpha \in \mathfrak{F}_1$. We let $\alpha > \beta$ if $H_\alpha \subset H_\beta$. Since $\mathfrak{F}$ is compact, the net $\{H_\alpha\}$, $\alpha \in J$, has

an adherent point K . It follows from Lemma 4 of [1] that $H \subseteq K$. Suppose that $H \neq K$ and $x \in K \setminus H$. We take a compact neighborhood V of the point x that is disjoint from H . Let us observe that $H_\alpha \cap V \neq \emptyset$ for any $\alpha \in J$, otherwise there would exist a $\beta \in J$ such that $H_\alpha \not\subseteq D_2(V)$ for any $\alpha > \beta$. The centered system of closed sets $\{H_\alpha \cap V\}$, $\alpha \in J$, has a common point $y \in V \cap H$ since V is compact. This contradicts the assumption that $V \cap H = \emptyset$. Therefore $K = H$. Since $n\mathfrak{R}(G)$ is closed in the E -topology, it follows that H is noncompact and we can directly use Corollary 1 of Lemma 2.

Remark 1. If a noncompact subgroup H is a limit point of a sequence of noncompact subgroups $\{H_n\}$, then H is a limit point for those subgroups of the sequence which are contained in it. Indeed, on assuming the contrary, we can suppose that $H_n \setminus H \neq \emptyset$ for any natural number n . Let U be a compact neighborhood of the identity. Using the noncompactness of $H_n \setminus H$, we can easily construct a sequence $\{x_n\}$ of elements of G such that $x_n \in H_n \setminus H$ and $x_i U \cap x_j U = \emptyset$ for $i \neq j$. The set $X = \{x_1, \dots, x_n, \dots\}$ is closed; however the neighborhood $D_1(G \setminus X)$ of the subgroup H does not contain any subgroup H_n .

LEMMA 3. Let H and X be, respectively, a noncompact and a compact subgroup of a topological group G . If $\mathfrak{H} = \{x^{-1}Hx, x \in X\}$ is contained in a compact subset of $\mathfrak{L}_E(G)$, then it is finite.

Proof. At first, we show that $\mathfrak{H}$ is closed in $\mathfrak{L}_E(G)$. Let F be an adherent point of $\mathfrak{H}$ and suppose that $\{x_\alpha^{-1}Hx_\alpha\}$, $\alpha \in J$, is a net of subgroups from $\mathfrak{H}$ that converges to F . The net $\{x_\alpha\}$, $\alpha \in J$, has a subnet $\{x_\beta\}$, $\beta \in J_1$, that converges to a certain $x \in X$. If $y = x^{-1}hx \in x^{-1}Hx$, but $y \notin F$, then by virtue of regularity of the space G there exist open sets U and V such that $F \subseteq U$, $y \in V$ and $U \cap V = \emptyset$. Then, starting from a certain $\beta_1 \in J_1$, we have $x_{\beta_1}^{-1}Hx_{\beta_1} \in D_1(U)$, and starting from a certain $\beta_2 \in J_1$, $x_{\beta_2}^{-1}hx_{\beta_2} \in V$, which contradicts the choice of U and V . Therefore $x^{-1}Hx \subseteq F$. Further, let $f \in F$ and $\{x_\gamma\}$, $\gamma \in J_2$, be a subnet of $\{x_\beta\}$, $\beta \in J_1$, such that $\{x_\gamma^{-1}Hx_\gamma\}$, $\gamma \in J_2$ and $h_\gamma \in H$ for each $\gamma \in J_2$, converges to f . Then $f = x^{-1}h_1x$, where $h_1 \in H$ and $f \in x^{-1}Hx$. Hence $F \subseteq x^{-1}Hx$ and, finally, $F = x^{-1}Hx$.

Let us suppose that $\mathfrak{H}$ is infinite and $y^{-1}Hy$ is a limit point of $\mathfrak{H}$. By virtue of Corollary 1 of Lemma 2, there exists a subgroup $z^{-1}Hz \in \mathfrak{H}$ such that $z^{-1}Hz \subseteq y^{-1}Hy$ and $z^{-1}Hz \neq y^{-1}Hy$. But then we can easily construct an infinite decreasing chain of subgroups from $\mathfrak{H}$, which contradicts Corollary 2 of Lemma 2.

COROLLARY. Let H be a noncompact subgroup of a totally disconnected group G and U be an open compact subgroup. If a compact subset F of $\mathfrak{L}_E(G)$ contains $\{x^{-1}Hx, x \in U\}$ and also all those subgroups of X which contain H , then H is open.

Proof. By Lemma 3, H is normalized by a certain open subgroup V of U . Therefore HF is a subgroup for each subgroup F of V , and the corollary follows immediately from Corollary 2 of Lemma 2.

2. Compactness of $\overline{\mathfrak{M}(G)}$, $\mathfrak{A}(G)$, and $\mathfrak{N}(G)$

The topological group G is not assumed to be locally compact in Theorems 1 and 2.

THEOREM 1. $\overline{\mathfrak{M}(G)}$ is compact in the E -topology if and only if $\mathfrak{L}_E(G)$ is compact.

Proof. Suppose that $\overline{\mathfrak{M}(G)}$ is compact and let $\{H_\alpha\}$, $\alpha \in J$, be an arbitrary set of subgroups of G . The net $\{H_\alpha\}$, $\alpha \in J$, has a subnet $\{H_\beta\}$, $\beta \in J_1$, that S -converges to a certain subgroup H (see [2]). We show that $\{H_\beta\}$, $\beta \in J_1$, converges to H in the E -topology. By virtue of the definitions of S -convergence and the E -topology, it is sufficient to verify that for each open set $U \supseteq H$ there exists a $\beta_0 \in J_1$ such that $H_\beta \subseteq U$ for all $\beta > \beta_0$.

Let us assume the contrary. Then there exist an open set U_0 that contains H and a subnet $\{H_\gamma\}$, $\gamma \in J_2$, of the net $\{H_\beta\}$, $\beta \in J_1$, such that H_γ is not contained in U_0 for any $\gamma \in J_2$. We select an element $h_\gamma \notin U_0$ from each subgroup H_γ . Since $\mathfrak{M}(G)$ is precompact, the net $\{\langle h_\gamma \rangle\}$, $\gamma \in J_2$, has a subnet $\{\langle h_\omega \rangle\}$, $\omega \in J_3$, that converges in the E-topology to a certain subgroup K . It follows from the definition of the E-topology that K is not contained in U_0 . Since each net of subgroups that is convergent in the E-topology is S-convergent, the net $\{\langle h_\omega \rangle\}$, $\omega \in J_3$, S-converges to K . Since $\langle h_\omega \rangle \subseteq H_\omega$ for each $\omega \in J_3$, it follows that $K \subseteq H$. We have obtained a contradiction with the fact that K is not contained in U_0 .

COROLLARY 1. $\mathfrak{M}(G)$ is compact in the E-topology if and only if $\mathfrak{L}_E(G)$ is compact.

Proof. Since $\mathfrak{M}(G) \subseteq \mathfrak{M}(\overline{G})$, necessity of the condition follows from Theorem 1. Sufficiency of the condition follows from Lemma 2 of [1].

COROLLARY 2. If G is locally compact and $\mathfrak{M}(G)$ is compact in the E-topology, then G is compact.

Proof. It follows from Theorem 1 that $\mathfrak{L}_E(G)$ is compact. If G is not compact, then by Lemma 10 of [1] it contains a subgroup A that is isomorphic to either C_{p^∞} or R_p . By Lemma 3 of [1], $A \in \mathfrak{M}(\overline{G})$. However $\mathfrak{M}(\overline{G}) = \mathfrak{M}(G)$. Therefore G is compact.

COROLLARY 3. If G is a totally disconnected locally compact group, then $\mathfrak{M}(G)$ is compact in the E-topology if and only if G is compact.

Proof. Compactness of G under the condition of compactness of $\mathfrak{M}(G)$ follows from Corollary 2. The converse follows from the fact that if a net $\{x_\alpha\}$, $\alpha \in J$, of elements of a totally disconnected compact group converges to x , then the net of subgroups $\{\langle x_\alpha \rangle\}$, $\alpha \in J$, converges in the E-topology to $\langle x \rangle$ (see the proof of Theorem 1 in [3]).

Remark 2. In general, compactness of G does not imply compactness of $\mathfrak{M}(G)$. As an example, we can take an arbitrary connected compact Abelian group of weight greater than the cardinality of the continuum.

A topological group G is said to be cramped (see [4]) if each subset of pairwise noncomparable subgroups from $\mathfrak{L}(G)$ is finite.

THEOREM 2. The lattice $\mathfrak{L}_E(G)$ of each compactly coverable cramped group G is compact.

Proof. By virtue of Theorem 1, it is sufficient to verify that $\mathfrak{M}(\overline{G})$ is compact. Let $\{\langle x_\alpha \rangle\}$, $\alpha \in J$, be an arbitrary net of monothetic subgroups of G . We select from it a subnet $\{\langle x_\beta \rangle\}$, $\beta \in J_1$, that S-converges to a certain subgroup X . If all subgroups $\langle x_\beta \rangle$ starting from a certain index are contained in X , then the net $\{\langle x_\beta \rangle\}$, $\beta \in J_1$, converges to X in the E-topology also. Otherwise there would exist a subnet $\{\langle x_\gamma \rangle\}$, $\gamma \in J_2$, such that no subgroup $\langle x_\gamma \rangle$ is contained in X . Since G is cramped and the net $\{\langle x_\gamma \rangle\}$, $\gamma \in J_2$, S-converges to X , we can find a $\gamma_0 \in J_2$ such that $\langle x_\gamma \rangle \subseteq \langle x_{\gamma_0} \rangle$ for all $\gamma > \gamma_0$. Since S-convergence is equivalent to convergence in the E-topology in the lattice $\mathfrak{L}(\langle x_{\gamma_0} \rangle)$ of the compact group $\langle x_{\gamma_0} \rangle$, the net $\{\langle x_\gamma \rangle\}$, $\gamma \in J_2$, converges to X in the E-topology.

COROLLARY. The lattice $\mathfrak{L}_E(G)$ of each locally compact cramped group G is compact.

Proof. This corollary follows from the fact that each locally compact cramped group can be compactly covered [4].

Remark 3. Since $\mathfrak{L}_E(G)$ is compact for each compact group G , compactness of the lattice $\mathfrak{L}_E(G)$ does not imply that G is cramped. However in the case of a noncompact locally compact group G , by virtue of [1, 4], compactness of $\mathfrak{L}_E(G)$ is equivalent to crampedness of G . It is interesting to prove this fact without using the description of locally compact groups with compact lattice $\mathfrak{L}_E(G)$ and also of locally compact cramped groups.

THEOREM 3. If G is a nilpotent group, then $\mathfrak{M}(G)$ is compact in the E-topology if and only if $\mathfrak{L}_E(G)$ is compact.

Proof. Since $\mathfrak{M}(G)$ is closed in $\mathfrak{L}_E(G)$ [1, Lemma 2], the sufficiency of the condition is obvious. Let us prove the necessity of the condition. We observe that if $\mathfrak{M}(G)$ is compact in the E-topology, then $\mathfrak{M}(H)$ and $\mathfrak{M}(G/N)$, where H is a central and N is a normal subgroup of G , are compact [1, Lemmas 2 and 5]. Therefore, it is natural to carry out the proof by induction on the degree of nilpotence of G . The theorem is obvious for Abelian groups. It follows from the results of [5] and Corollary 2 of Lemma 2 that G can be compactly covered. Further, we assume that G is not compact and Z is the center of G .

Let us suppose that G is a totally disconnected p -group and let us consider two cases.

1) G/Z is not compact. By the induction hypothesis and Theorem 4 of [1], G/Z is isomorphic to either C_{p^∞} or R_p . Since Z is central, it follows that G is Abelian.

2) G/Z is compact. Then Z is isomorphic to either C_{p^∞} or R_p , and, by Corollary 2 of Lemma 2, G/Z is finite. Let $a_1, \dots, a_n$ be representatives of cosets of G modulo Z and $Z \simeq C_{p^\infty}$. The subgroup $N = \langle a_1, \dots, a_n \rangle$ is finite and normal in Z ; moreover, $G = ZN$. Further, $G/Z \cap N \simeq (Z/Z \cap N) \times (N/Z \cap N)$. Since $Z/Z \cap N \simeq C_{p^\infty}$, and $N/Z \cap N$ is a nilpotent p -group, we necessarily have $N = Z \cap N$ and $G \simeq C_{p^\infty}$ (otherwise, the center of $G/Z \cap N$ would have a noncompact lattice of subgroups). The case $Z \simeq R_p$ is reduced to the one considered by factorization by an open compact subgroup of Z .

Thus, if G is totally disconnected, then a Sylow p -subgroup of G is either isomorphic to C_{p^∞} or R_p or it is compact. Using Corollary 2 of Lemma 2, we conclude that $\mathfrak{L}_E(G)$ is compact.

Let us assume that the connected component $G_0 \neq e$. Factoring G by a certain subgroup of G_0 , we can assume that $G_0 \simeq T$. By what we have proved above, $\mathfrak{L}_E(G/G_0)$ is compact. Let $G/G_0 \simeq C_{p_1^\infty} \times \dots \times C_{p_n^\infty} \times K$, $p_1, \dots, p_n$ be different prime numbers, K be finite, and the order of K be not divisible by $p_1, \dots, p_n$. Let M denote the inverse image in G of a subgroup isomorphic to $C_{p_1^\infty} \times \dots \times C_{p_n^\infty}$, and N denote the inverse image of a subgroup isomorphic to K . Since G_0 is central and M/G_0 is the union of a sequence of cyclic subgroups, it follows that M is Abelian. Further, $N = G_0 \cup G_0 b_1 \cup \dots \cup G_0 b_m$, where, since G_0 is divisible and central, we can assume that $b_1, \dots, b_m$ are elements of finite order. Since N has only a finite number of elements of order n for each natural number n , the elements $b_1, \dots, b_m$ are contained in a finite normal subgroup N_1 of G and $G = MN_1$. Since M normalizes N_1 and does not have any subgroup of finite index, it follows that M centralizes N_1 . Therefore, M is central in G and $\mathfrak{L}_E(M) \subseteq \mathfrak{R}(G)$. However $G_0 \subseteq M$, and therefore $\mathfrak{L}_E(M)$ is not compact [1, Theorem 4]; we have obtained a contradiction with the compactness of $\mathfrak{R}(G)$. The case $G/G_0 \simeq R_p \times K$ is considered in the same manner.

Remark 4. Theorem 3 cannot be extended either to solvable or to ZA-group. Let G be the semidirect product of a discrete quasicyclic group F with a group of second order. Since each normal subgroup of G is either contained in F or coincides with G , it follows that $\mathfrak{R}(G)$ is compact in the E -topology. However, $\mathfrak{L}_E(G)$ is not compact.

3. Compactness of $\mathfrak{R}(G)$, $\overline{\mathfrak{R}(G)}$, and $n\mathfrak{R}(G)$

THEOREM 4. $\overline{\mathfrak{R}(G)}$ is compact in the E -topology if and only if G is the extension of an EK-group by means of a pure group.

Proof. Let $\overline{\mathfrak{R}(G)}$ be compact and $K(G)$ be the set of compact elements of G . By Lemma 2 of [1] and Theorem 1, each compactly coverable subgroup of G is an EK-group. By virtue of Theorem 3 of [6], it is sufficient to show that $K(G)$ is a subgroup. Let us consider the following two cases:

1) $\overline{\mathfrak{R}(G) \cap \mathfrak{R}(G)} \subseteq \mathfrak{R}(G)$. By Lemma 1, $K(G)$ is contained in a compact set and, by virtue of [7], in a compact normal subgroup.

2) $\overline{\mathfrak{R}(G) \cap \mathfrak{R}(G)}$ contains a noncompact subgroup H . It follows from Corollary 1 of Lemma 2 that the set of compact elements is dense in H . By Lemma 2 of [1], H is Abelian. Therefore, H can be compactly covered and is, consequently, an EK-group. Again there are two possibilities:

2a) H contains a subgroup $A \simeq C_{p^\infty}$. Let $x \in K(G)$. By Lemma 3, H is normalized by a subgroup X of $\overline{\langle x \rangle}$ of finite index. Since $\overline{\langle H, X \rangle} = HX$ can be compactly covered, it follows by Theorem 4 of [1] that x has a finite order n . Let us set $\mathfrak{H} = \{g^{-1} \langle x \rangle g, g \in G\}$. It follows from the proof of Lemma 8 of [1] that each adherent point of $\mathfrak{H}$ contains at most n elements. Therefore $\mathfrak{H} \subseteq \mathfrak{R}(G)$ and, by Lemma 1, $\overline{\bigcup_{g \in G} g^{-1} x g}$ is compact. Using [7], we conclude that $K(G)$ is a subgroup.

2b) H contains a subgroup $B \simeq R_p$. Let us suppose that $g^{-1}Bg \neq B$ for a certain $g \in G$. By Lemma 3, a nontrivial subgroup Y of $g^{-1}Bg$ normalizes B . Since B can be compactly covered, Y is an EK-group. By Theorem 4 of [1], $Y \subseteq B$ and $g^{-1}Bg \cap B \neq e$. If $g^{-1}Bg \neq B$, then $\overline{\langle g^{-1}Bg, B \rangle} / (g^{-1}Bg \cap B)$ contains two different subgroups that are isomorphic to C_{p^∞} , which contradicts 2a). Therefore B is normal in G . Let F be an open compact subgroup of B . Since each compact element of G , acting by conjugations on B , fixes F , F is normal in $\overline{\langle K(G) \rangle}$. Since $\overline{\langle K(G) \rangle} / F$ contains a subgroup that is isomorphic to C_{p^∞} , it is a discrete EK-group. Hence

it follows immediately that $\overline{\langle K(G) \rangle} = K(G)$, and necessity of the condition is proved.

Sufficiency of the condition follows from Lemma 5 of [1].

COROLLARY. $\mathfrak{R}(G)$ is compact in the E-topology if and only if G is the extension of a compact group by means of a pure group.

Proof. This corollary follows from the description of an EK-group and Theorem 4.

A topological group G with the set $n\mathfrak{R}(G)$ compact in the E-topology is called an ES-group. Since $n\mathfrak{R}(G)$ is closed in $\mathfrak{L}_E(G)$, each EK-group is an ES-group. Another example of ES-groups is the Schmidt topological group, i.e., a noncompact group, all proper subgroups of which are compact. Moreover, it follows from Corollary 2 of Lemma 2 that each noncompact ES-group contains a Schmidt subgroup. Therefore description of an ES-group is possible only up to a Schmidt group. Let us recall that Abelian Schmidt groups are exhausted by C_{p^∞} and R_p .

THEOREM 5. The following types of groups exhaust the class of the ES-groups that do not contain non-Abelian Schmidt subgroups:

- 1) compact groups;
- 2) $C_{p_1^\infty} \times \dots \times C_{p_n^\infty} \times K$, where $p_1, \dots, p_n$ are different prime numbers, K is finite, and the order of K is not divisible by $p_1, \dots, p_n$;
- 3) finite extensions of C_{p^∞} and R_p .

THEOREM 6. If a discrete ES-group G contains a non-Abelian Schmidt subgroup H , then H is normal and G/H is an EK-group.

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